Mineral Detection From Satellite Image Using Explainable AI (XAI)

Presented By

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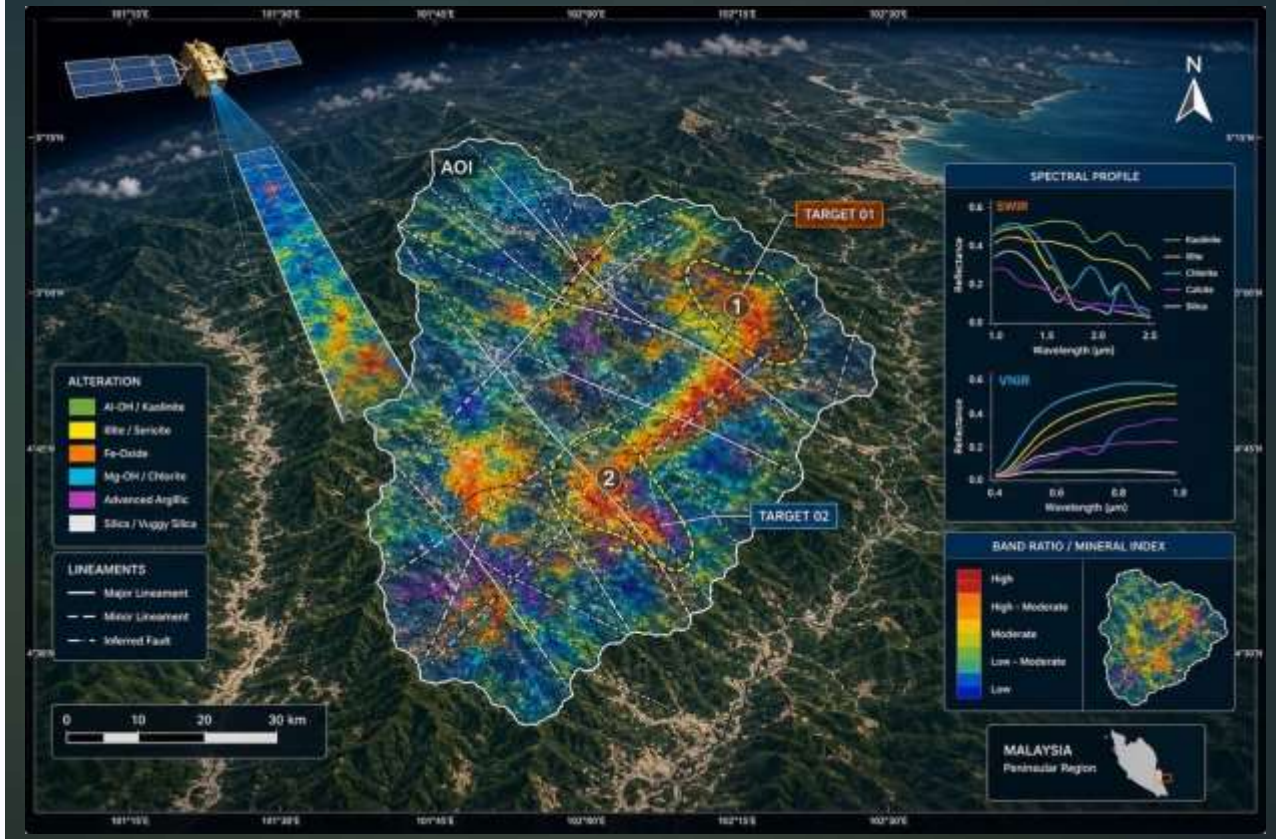
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Alunite mapping over the Cuprite dataset, with future neural-network abundance estimation.

What is Hyperspectral Imagery?

Hyperspectral sensors capture hundreds of narrow spectral bands per pixel, generating unique **spectral signatures** for each material. These signatures enable precise **mineral identification** from space — without physical sampling.



Introduction, Problem Statement & Motivation

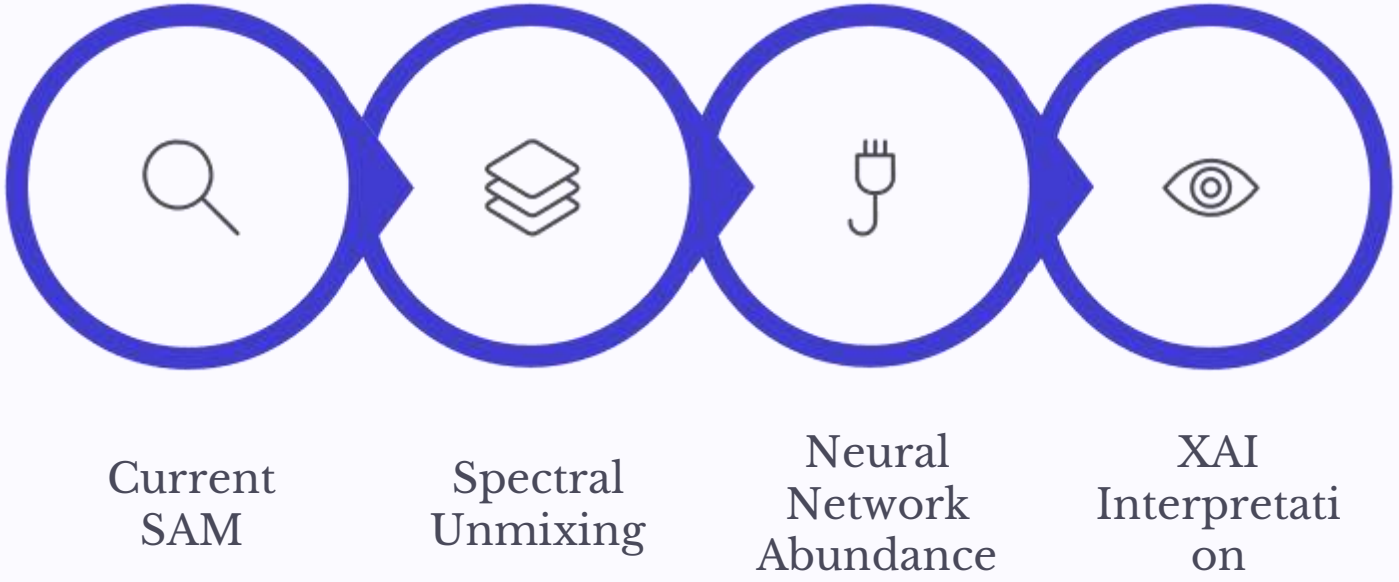
Problem Statement

> How can we estimate mineral abundance from mixed pixels in hyperspectral imagery?

Key Challenges

- Geological exploration requires precise, spatially accurate mineral mapping
- Mixed pixels contain contributions from multiple minerals simultaneously
- Binary detection alone is insufficient — continuous abundance estimation is needed

Proposed Approach



Literature Review & Related Work

Four complementary approaches to mineral detection and abundance estimation:

SAM — Spectral Angle Mapper

Compares pixel spectra to references using spectral angle. Interpretable similarity measure and foundational method.

Matched Filtering

Compares target and background spectral response to produce a target score.

Linear Spectral Unmixing

Models pixels as mixtures of endmember spectra to estimate abundance fractions.

Deep Learning Unmixing

Uses neural networks to learn nonlinear spectral relationships for abundance estimation.

Dataset Understanding: Cuprite

Cuprite Dataset Dimensions

47,500	188	12
Total Pixels	Selected Bands	Endmembers
250 × 190 spatial grid	From 224 original wavelength samples	Reference spectra in spectral library

Target Mineral

Alunite #1 — selected as the primary endmember for SAM-based detection. The Cuprite scene is a well-established AVIRIS benchmark for hyperspectral algorithm validation.

Processing Pipeline

Dataset inspection → Band identification → Alunite endmember selection

Preprocessing & Band Alignment

Ensuring spectral dimension consistency between the image cube and the Alunite reference spectrum:

1 Load Cuprite .mat data

Import raw hyperspectral image cube from MATLAB format.

2 Extract 188 image bands

Isolate the pre-selected valid spectral bands, discarding water-absorption regions.

3 Load 224 endmember wavelength samples

Read the full spectral library reference at native resolution.

4 Band selection / slctBnds alignment

Map the 224 endmember bands to the 188 image bands via index alignment.

SAM Methodology

Spectral Angle Mapper Principle

SAM treats each pixel spectrum and the reference endmember as **vectors in n-dimensional spectral space**. The angular distance between them measures spectral similarity — independent of illumination magnitude.

- **Pixel spectrum x** vs **Alunite spectrum s**
- Smaller angle → greater similarity → stronger Alunite signature
- Larger angle → lower similarity → pixel is unlikely Alunite

Reference: Kruse et al., *Remote Sensing of Environment*, 1993.

Mathematical Formulation

$$\theta = \cos^{-1}\left(\frac{\mathbf{x}\ \mathbf{s}}{\|\mathbf{x}\|\ \|\mathbf{s}\|}\right)$$

where θ is the spectral angle in radians, $\mathbf{x}$ is the pixel spectrum vector, and $\mathbf{s}$ is the Alunite reference spectrum vector.

Key Properties

- Range: $0 \leq \theta \leq \pi$
- Scale-invariant — insensitive to brightness variation
- Fully interpretable — no black-box components

Proposed Pipeline & Detection Threshold

Decision Threshold Logic

✓ Detected	✗ Not Detected
$\theta \leq 0.15$ rad	$\theta > 0.15$ rad
Pixel classified as containing Alunite	Pixel classified as non-Alunite

Output

The pipeline produces a **Binary Alunite Detection Map** — a 250 × 190 spatial grid where each pixel is labeled as detected or not detected based on its spectral angle to the Alunite #1 endmember.

The threshold of **0.15 rad** is applied uniformly across all 47,500 pixels. Tightening the threshold reduces false positives; relaxing it increases detection sensitivity.

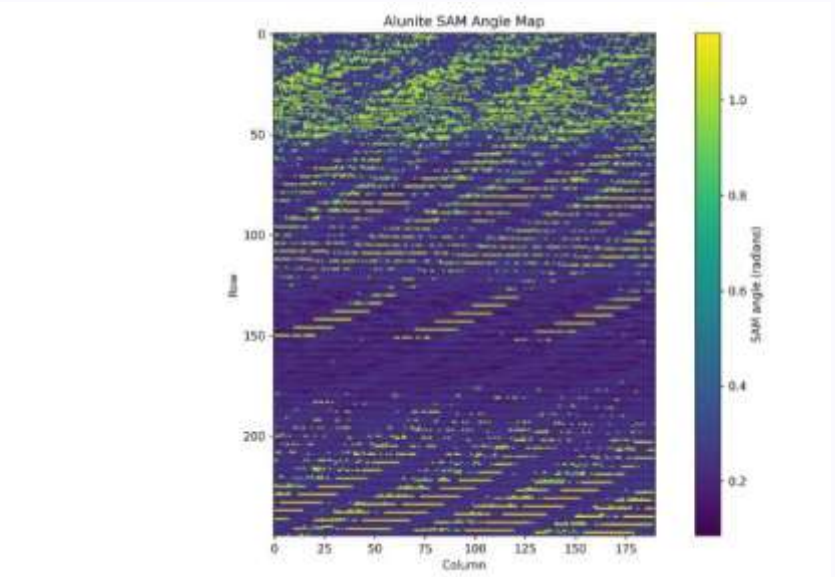
Results & Analysis

Detection Statistics

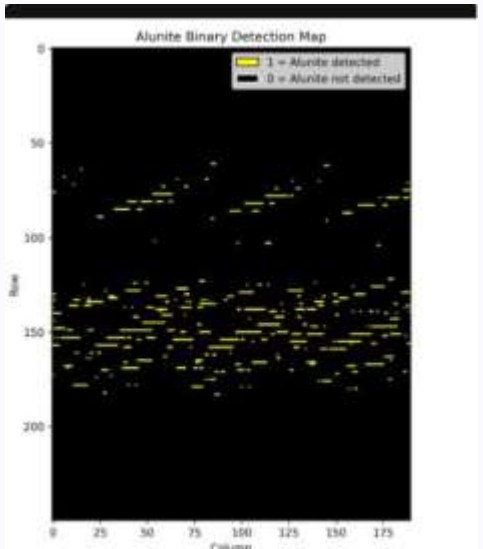
Metric	Value
Total pixels processed	47,500
Detected (θ ≤ 0.15 rad)	852
Not detected (θ > 0.15 rad)	46,648
Threshold applied	0.15 rad
Detection percentage	1.7937%

$$\frac{852}{47{,}500} \times 100 = 1.793684\%$$

SAM Angle Map



Binary Detection Map



Current Approach: SAM Detection

Question

Is Alunite *present*?

Process

Pixel → SAM angle → Binary threshold

Output

Yes / No per pixel

Future Work

Neural Network Abundance Estimation

Explainable AI Integration

Future XAI methods (e.g., SHAP, LIME, saliency maps) will interpret neural-network predictions and provide transparent, band-level explanations for abundance estimates — making results accessible to domain geologists.

Implementation& References

Implementation Stack

Language

Python

Libraries

NumPy · SciPy · Matplotlib

Modules

main.py · load_cuprite.py · preprocessing.py · sam.py · detection.py · visualization.py

References

[1] Kruse, F. A. et al., "The Spectral Image Processing System (SIPS)—Interactive Visualization and Analysis of Imaging Spectrometer Data," *Remote Sensing of Environment*, 1993.

[2] Bioucas-Dias, J. M. et al., "Hyperspectral Unmixing Overview: Geometrical, Statistical, and Sparse Regression-Based Approaches," *IEEE JSTARS*, 2012.

[3] Qi, L. et al., "Deep spectral convolution network for hyperspectral image unmixing with spectral library," 2020.

[4] Zhou, C. & Rodrigues, M. R. D., "ADMM-Based Hyperspectral Unmixing Networks for Abundance and Endmember Estimation," *IEEE TGRS*, 2021.

[5] Cuprite hyperspectral dataset / AVIRIS and associated spectral-library references.